

# Benjamin Yao

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## EDUCATION

**The University of Texas at Austin**, Austin, TX

Aug 2018 – May 2022

*Bachelor of Science in Physics and Astronomy*

GPA: 3.9 (out of 4.0)

## PROJECTS

- **Travel Application:** Developed a scalable full-stack application inspired by Airbnb/Zillow that utilizes the Google Places API and a self-made REST API to fetch rental home data from a MongoDB sample database. This application was containerized and hosted on AWS with application load balancing and an SSL/TLS Certificate. Built with Python, JavaScript, HTML/CSS, React, Django, Bootstrap5.x. Database: MongoDB. See GitHub for demo.
- **Finance Application:** Developed a Flask web app designed to mimic buying and selling shares of stocks. Used the IEX API to get stock quotes and created a back-end database to store “purchases”. Also created a second database that was used for authentication and account creation. Built with Python, Flask, HTML, Bootstrap5.x. Database: SQLite. See GitHub for demo.
- **Email Application:** Developed an email application. Frontend uses Django API. Built with Python, Django, JavaScript, HTML, Bootstrap5.x. Database: SQLite. See GitHub for demo.
- **E-Commerce Application:** Developed an eBay-like website for auction sales. Built with Python, Django, HTML, Bootstrap5.x. Database: SQLite. See GitHub for demo.
- **Discord Music Bot:** Developed a bot that responds to commands and can play a queue of music through a voice channel on Discord.com. Built with Python, this application was containerized and hosted on Fly.io to be available 24/7 on any server.
- **Restaurant Application:** Developed a burger building and ordering web app (<https://react-burger-d3798.web.app/>). Built with JavaScript, React, HTML, Bootstrap5.x.
- **Stochastic Gravitational Wave Machine Learning Project:** Developed and trained a neural network to learn to differentiate stochastic gravitational waves from background noise using publicly available data from the H1 and L1 detectors at LIGO.

## EXPERIENCE

**Comcast**, Philadelphia, PA

Jun 2022 – Present

*Auditor, Technology*

- Conducted audits across company technologies (e.g., physical, network, application, system, data). Assisted in identifying technologies that were not operating in accordance with company policies and procedures to minimize potential business risks.
- Assisted in developing audit scope and overall planning efforts. Performed thorough inquiry and data analysis to understand security and technology operations and assess risk.
- Identified remediation tactics after evaluating records, documents, methods, policies, costs, and other factors to determine if areas were conducting operations in accordance with established policies and procedures.
- Prepared clear and well-organized audit work papers documenting root-cause, work performed, and investigation summaries. Ensured all controls were tested and that all significant risks were addressed.
- Formulated appropriate conclusions and documented findings based on testing results. Combined knowledge of company operations with testing results to determine overall control effectiveness.

**Comcast**, Philadelphia, PA

Jun 2021 – Aug 2021

*Software Engineer Intern*

- Assisted in the development of the TES Connector web application and mobile application. The TES Connector mobile application was used to scan different IT assets (servers, routers, etc.) and inventory them in a database. The TES Connector web application was the frontend site used to display the properties of IT assets that were scanned with the mobile application.
- Wrote scripts to update the backend database and created end user documentation for the TES Connector web application.
- Script was designed to pull asset data from CSV files and add or update the properties of IT assets in the database accordingly. Project removed the need for a manual upload of items from CSV files to the database. (Python, SQL)

## TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, R, Java, MatLab, and Mathematica
- **Frameworks:** React, Flask, Django, Bootstrap, SASS
- **Tools:** Docker, AWS, Git, MongoDB, SQLite, Postgres